

TUMNIN, Russian Federation

WMO: 316830

Lat: 49.6708N

Lon: 140.1013E

Elev: 58

StdP: 100.63

Time Zone: 10.00 (E10)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.2	(c) -33.0	(d) -37.6	(e) 0.1	(f) -34.8	(g) -35.3	(h) 0.1	(i) -32.7	(j) 6.9	(k) -13.3	(l) 5.0	(m) -11.5	(n) 0.0	(o) 0	(p) 0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	28.9	19.9	26.8	19.3	24.8	18.3	21.5	26.8	20.4	25.1	19.3	23.4	2.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.5	23.9	18.7	13.6	22.8	17.7	12.7	21.6	63.3	27.0	59.2	25.3	55.4	23.5	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.3	(b) 4.3	(c) 3.5	(e) -38.7	(f) 34.0	(g) 2.4	(h) 2.3	(i) -40.4	(j) 35.6	(k) -41.8	(l) 37.0	(m) -43.2	(n) 38.3	(o) -44.9	(p) 39.9		
(a) 5.3	(b) 4.3	(c) 3.5	(e) -38.8	(f) 24.5	(g) 2.3	(h) 2.4	(i) -40.4	(j) 26.3	(k) -41.7	(l) 27.7	(m) -43.0	(n) 29.1	(o) -44.6	(p) 30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.7	-20.0	-15.9	-7.5	1.8	8.5	14.1	18.0	18.1	12.4	3.5	-8.3	-17.7		
	DBStd	14.25	6.40	5.31	5.16	3.60	3.71	3.88	3.01	2.81	3.62	3.83	6.21	6.21		
	HDD10.0	4152	931	724	543	247	74	6	0	0	18	203	549	857		
	HDD18.3	6522	1189	957	801	497	306	135	43	39	181	459	799	1116		
	CDD10.0	748	0	0	0	1	27	131	248	251	89	2	0	0		
	CDD18.3	76	0	0	0	0	1	9	32	32	2	0	0	0		
	CDH23.3	773	0	0	0	1	29	202	317	203	22	0	0	0		
	CDH26.7	198	0	0	0	0	8	62	91	34	3	0	0	0		
(14) Wind	WSAvg	1.1	0.9	1.0	1.2	1.3	1.4	1.1	1.0	0.8	0.9	1.2	1.1	1.0		
(15) Precipitation	PrecAvg	628	21	18	21	42	69	61	85	97	88	56	33	34		
	PrecMax	828	60	97	57	82	129	155	204	220	200	141	70	115		
	PrecMin	394	1	1	5	4	16	19	4	16	34	8	2	2		
	PrecStd	97	16	23	13	21	31	29	56	48	47	39	23	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.6	1.3	9.2	20.0	27.2	32.1	32.4	29.8	25.8	18.9	7.2	-0.4		
		MCWB	-2.8	-2.4	3.4	10.0	14.4	19.3	21.4	21.9	17.7	11.6	3.0	-1.2		
	2%	DB	-3.9	-2.2	5.1	14.8	22.8	28.1	29.3	27.5	22.7	15.1	4.5	-2.3		
		MCWB	-5.3	-4.3	0.7	7.3	12.7	18.9	20.5	20.4	16.4	9.0	2.0	-3.5		
	5%	DB	-6.2	-4.5	2.9	11.3	19.5	25.2	26.9	25.6	20.4	12.3	2.6	-4.8		
		MCWB	-7.5	-6.3	-0.6	5.2	11.6	17.7	19.6	19.6	15.4	7.7	0.4	-5.8		
	10%	DB	-9.2	-6.8	0.8	8.6	16.7	22.7	24.6	23.8	18.3	10.1	0.8	-7.4		
		MCWB	-10.5	-8.4	-1.8	3.6	10.2	16.1	18.7	18.8	14.4	6.7	-1.1	-8.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.4	-1.9	3.8	10.7	16.2	21.7	23.0	23.1	19.5	12.4	4.2	-0.8		
		MCDB	-1.7	0.6	8.6	19.2	25.0	29.8	28.8	28.7	23.5	16.2	6.3	-0.5		
	2%	WB	-5.2	-4.3	1.1	7.7	13.9	19.8	21.6	21.5	17.7	10.2	2.3	-3.7		
		MCDB	-4.1	-2.1	4.6	13.8	21.2	25.7	27.6	25.7	20.8	13.4	3.7	-2.7		
	5%	WB	-7.5	-6.2	-0.3	5.6	12.2	18.3	20.4	20.5	16.4	8.6	0.7	-5.8		
		MCDB	-6.3	-4.5	2.3	10.7	18.4	24.0	25.8	24.2	19.2	11.5	2.4	-4.7		
	10%	WB	-10.6	-8.3	-1.5	3.9	10.7	16.7	19.2	19.4	15.1	7.0	-0.9	-8.6		
		MCDB	-9.3	-6.8	0.8	8.1	16.1	22.1	23.7	22.7	17.8	9.7	0.7	-7.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	15.9	14.4	11.5	13.0	13.4	11.2	10.2	11.7	12.3	11.2	12.2		
		MCDBR	9.8	16.3	16.6	18.3	19.9	19.4	16.6	14.1	15.0	17.0	10.8	9.3		
		MCWBR	8.9	13.9	12.0	10.7	10.8	10.6	8.7	7.2	8.5	10.8	8.0	8.6		
		MCDBR	8.9	15.6	14.1	17.3	17.7	17.3	14.6	12.2	12.3	15.2	9.4	8.9		
	5% WB	MCWBR	7.9	13.3	10.1	10.4	10.0	9.9	8.3	7.0	7.3	10.2	7.1	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.321	0.395	0.491	0.547	0.516	0.485	0.430	0.387	0.398	0.348	0.291		
	taud		2.098	1.988	1.841	1.775	1.801	1.978	2.130	2.308	2.355	2.128	2.079	2.082		
	Ebn at Noon		735	789	782	748	727	757	776	807	805	695	633	655		
	Edh at Noon		83	120	168	201	205	173	147	117	101	106	86	74		
(44) All-Sky Solar Radiation	RadAvg		1.46	2.61	3.88	4.45	4.84	5.24	4.83	4.25	3.53	2.27	1.35	1.06		
	RadStd		0.12	0.14	0.27	0.46	0.39	0.55	0.49	0.38	0.32	0.15	0.12	0.12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+0.35	+1.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page